Lab 5 Code

October 2, 2025

[20]:
```
from datascience import *
```

[2]:
```
credit_score = make_array(800)
```

[3]:
```
if credit_score>=750:
        credit_category='Excellent'
elif credit_score>=700:
        credit_category='Good'
else:
        credit_category='Other'
print(credit_category)
```

```
Excellent
```

[4]:
```
if credit_score>=750:
        credit_category='Excellent'
if credit_score>=700:
        credit_category='Good'
if credit_score<700:
        credit_category='Other'
print(credit_category)
```

```
Good
```

[5]:
```
#Using 'if' statements only without specified bounds will give an incorrect␣
 ↪result
```

[6]:
```
import numpy
```

[7]:
```
numpy.random.uniform(1.8, 8.2)
```

[7]:
```
7.760002264814144
```

[8]:
```
Coffee = Table().with_columns('Wait Time',numpy.random.uniform(1.8,8.2,1000000))
```

[9]:
```
Coffee
```

[9]:
```
Wait Time
2.66956
```

```
4.26102
5.03717
6.34639
2.95715
8.04305
3.18329
5.77065
2.14071
6.59028
… (999990 rows omitted)
```

[10]:
```
SumValue = 0
for n in Coffee[0]:
    if n >= 2.5 and n <= 6:
        SumValue = SumValue + 1
print(SumValue/Coffee.num_rows)
```

```
0.547183
```

[11]:
```
SumValue = 0
for n in Coffee[0]:
    if n >= 4:
        SumValue = SumValue + 1
print(SumValue/Coffee.num_rows)
```

```
0.656192
```

[12]:
```
SumValue = 0
for n in Coffee[0]:
    if n < 4:
        SumValue = SumValue + 1
print(SumValue/Coffee.num_rows)
```

```
0.343808
```

[13]:
```
sum(Coffee[0])/Coffee.num_rows
```

[13]:
```
4.9986041285221106
```

[14]:
```
numpy.random.normal(98.2, 0.4)
```

[14]:
```
98.20260778449388
```

[15]:
```
Temps = Table().with_columns('Body Temp',numpy.random.normal(98.2,0.4,1000000))
```

[16]:
```
Temps
```

[16]:
```
Body Temp
98.4206
```

```
98.6538
98.0724
97.8776
98.4618
97.8069
97.5626
98.2944
98.3274
98.5587
… (999990 rows omitted)
```

[17]:
```
SumValue = 0
for n in Temps[0]:
    if n >= 98.8:
        SumValue = SumValue + 1
print(SumValue/Temps.num_rows)
```

```
0.067065
```

[18]:
```
SumValue = 0
for n in Temps[0]:
    if n >= 97.5 and n <= 99:
        SumValue = SumValue + 1
print(SumValue/Temps.num_rows)
```

```
0.936984
```

[19]:
```
sum(Temps[0])/Temps.num_rows
```

[19]:
```
98.200564337812608
```

[ ]:
```

```